

Performance and enterprise grids

@defer · OnPush · MatTable · paginator TMS on weak tablets

TMS command center

```
@defer (on viewport) {  
  <tms-heavy-analytics-chart />  
} @placeholder { ... } @loading { ... }
```

Ship less before first paint

Session goals

By the end of this session, you will:

- Explain why a monolithic main.js blocks first interaction on slow networks (spine)
- Defer below-the-fold TMS widgets with `@defer` separate compile-time chunks (spine)
- Use triggers, `@placeholder`, `@loading`, `@error` minimum delay to avoid flicker (spine)
- Apply `ChangeDetectionStrategy.OnPush` so 5,000-row grids do not repaint on every keystroke (spine)
- Contrast plain HTML tables with `MatTable` sort, paginate, accessibility trade-offs (spine)
- Avoid Session 2 traps: wrong defer, `OnPush` without signals, missing animations provider (mistakes table)

The 4MB bundle problem

Angular SPA navigation feels instant once the app is running but the initial load is the bottleneck.

Angular bundles everything the route needs into a large main.js chunk. The browser downloads the whole app before the first button is trustworthy.

Goal: do not send code until the user actually needs that UI (spine).

Breaking the bundle with @defer

- Route lazy load (loadComponent) is not enough heavy pieces inside a page still ship in the parent chunk
- @defer splits wrapped components into their own .js chunk at compile time (Angular 22 spine)
- Typical trigger: on viewport download when the user scrolls the analytics region into view

Sequence (spine mermaid in prose)

User opens dashboard → header/sidebar from main.js render immediately → user scrolls below the fold → @defer triggers network fetch → @placeholder / @loading / then heavy chart component (spine).

The @defer block structure

Template (spine)

```
@defer (on viewport; prefetch on idle(500)) {  
  
  <tms-heavy-analytics-chart />  
  
} @placeholder {  
  
  <div>Scroll down to load chart</div>  
  
} @loading (minimum 500ms) {  
  
  <div>Downloading chart code...</div>  
  
} @error {  
  
  <div>Failed to fetch module.</div>  
  
}
```

- `@defer` master block; compiler splits deferred subtree into its own chunk
- `on viewport` trigger when the block enters the visible area
- `@placeholder` instant shell before trigger; `@loading` during fetch; `minimum 500ms` reduces flicker (spine)
- `@error` honest failure surface if the chunk never arrives

Change detection: the silent killer

OnPush (spine pattern)

```
@Component({  
  selector: 'tms-enrollment-row',  
  changeDetection: ChangeDetectionStrategy.OnPush, // Angular 22 default  
  template: `<td>{{ data().name }}</td>`  
})
```

- Angular 22 default: OnPush is now the CLI default legacy check-all renamed Eager (deprecated) (spine)
- OnPush: check this component when @Input references change or a signal() it reads emits
- CoTBE pairing: OnPush + signal() for row data junior default vs senior discipline (spine)

Angular Material data tables

HTML table	MatTable
Fast initial build	Learning curve for API
No built-in sort	matSort built in
No built-in pagination	matPaginator built in
Accessibility is manual	ARIA + keyboard navigation built in

MatTable anatomy (spine trimmed)

```
<table mat-table [dataSource]="dataSource" matSort>

  <ng-container matColumnDef="studentName">

    <th mat-header-cell *matHeaderCellDef mat-sort-header>Student</th>

    <td mat-cell *matCellDef="let element">{{element.studentName}}</td>

  </ng-container>

  <tr mat-header-row *matHeaderRowDef="displayedColumns"></tr>

  <tr mat-row *matRowDef="let row; columns: displayedColumns;"></tr>

</table>

<mat-paginator [pageSizeOptions]="[10, 25, 50]"></mat-paginator>
```

Common mistakes Session 2

Mistake	Symptom	Fix
Wrong @defer usage	First load takes 10 seconds (spine symptom)	Wrap heavy components in @defer (on viewport); verify chunk generation in the build console (spine)
OnPush without reactive inputs	TS updates but template stays stale	Drive UI with signal() (or async pipe) so OnPush knows when to run (spine)
Forgetting animations	Sort arrows do not animate; overlays misbehave	provideAnimations() in app.config.ts (or legacy BrowserAnimationsModule) (spine)

Session 2 summary

- @defer compile-time split + viewport trigger for below-the-fold TMS UI
- OnPush + signals stop grid-wide repaint storms on instructor search
- MatTable enterprise sort/paginate/a11y baseline for enrollment lists

Next: Session 3 defensive RxJS and live sync

exhaustMap, duplicate-click defense, SignalR for honest seat counts TMS money and trust.